

Workforce Planning Application Specification Document

1. Overview

This document provides the functional and structural specifications required for the development of the Workforce Planning Software Application. The application is designed to support Capability Analysis, Role Analysis & Talent Segmentation, Demand Planning, and Budget Tracking workflows.

It is based on the provided Excel template, ERD diagram, and UI wireframe prototype.

2. Included Reference Files

Item	Description	Link / Location
Workforce Planning Template	Source Excel with structure and calculation logic	Download Excel Template
ERD Diagram	Data model structure in PNG format	Download ERD Diagram
Public Wireframe Prototype	Clickable user interface model	Import this ZIP in CodeSandbox → Share → Copy Link: Download wireframe ZIP

Note: The wireframe link can be shared by deploying the included React wireframe to Vercel, Netlify, or CodeSandbox. Replace placeholder once hosted.

3. Application Modules

A. Capability Analysis

- Create and manage capabilities (Primary/Secondary).
- Define descriptions, focus areas, KPIs, and associated skills.
- Link capabilities to roles.

B. Role Analysis & Talent Segmentation

- Create and categorize roles under capabilities.
- Assign focus areas, KPIs, and skills.
- Score **Criticality** and **Scarcity**.
- System computes quadrant classification (A/B/C/D) and recommended action.

C. Demand Planning

- Define planning period (monthly or quarterly).
- Enter required headcount by role and time period.
- Compute total cost based on cost model.
- Compare with manual budget input and calculate variance.

D. Budget Tracking & Reporting

- Summaries by capability, role, employment type, and period.
- Export and approval-ready reports.

4. Data Model Summary (ERD Reference)

The data model supports modular and scalable planning with normalization for roles, capabilities, planning periods, and cost computation.

Key tables include: - Capability - Role - Skill - KPI - SubActivity - PlanningHeader - DemandLine - DemandLinePeriod - ConfigRoleEvaluation

Refer to ERD for relationships and keys.

5. Key Business Rules

- Capability and Role names must be unique.
- Quadrant classification is entirely formula-driven.
- Each Demand Line supports multiple periods with computed totals.
- Budget variance is calculated as: **Total Demand Cost – Manual Budget.**

6. User Roles & Permissions

Role	Access Scope
Administrator	Full system configuration and master data setup
Planner / HR COE	Create and edit capabilities, roles, and demand plans
Finance Reviewer	Manage budgets, approve plans
Viewer	Read-only dashboards & exports

7. Technology Recommendations

- **Frontend:** React / Next.js + shadcn/ui design system
 - **Backend:** Node.js (NestJS) or Python (FastAPI)
 - **Database:** PostgreSQL
 - **Authentication:** SSO (SAML/OIDC) if required
 - **Deployment:** Cloud (AWS, Azure, GCP, or Vercel for UI hosting)
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8. Next Steps

Step	Owner	Status
Confirm hosting of Wireframe Link	Sender / UI Developer	Pending
Finalize Scope & Phase Breakdown	Stakeholders	Pending
Begin UI & Data Model Implementation	Engineering Team	Pending

9. Contact

For clarifications or implementation support, contact: **[Insert Name / Email / Team Here]**.

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